A Lean-verified offset construction for the Erdős–Straus equation

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Abstract

We give a self-contained formulation of an offset construction for the Erdős–Straus equation

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

The construction packages the classical divisor split into a congruence condition indexed by a prime $p \equiv 3 \pmod 4$. Its full positive-integer statement is formalised in Lean 4 with Mathlib: the development constructs the denominators by natural-number division, proves the divisions exact and the denominators positive, and concludes the equality in $\mathbb{Q}$. We also formalise a fixed-divisor lifting theorem and three groups of explicit infinite families, including the progressions arising from $p = 3$ and $d_1 = 2, 5$. We make no claim toward resolving the Erdős–Straus conjecture; the underlying divisor split is classical.

1 The divisor split

The Erdős–Straus conjecture asks whether, for every integer $n \geq 2$, there exist $x, y, z \in \mathbb{N}_{>0}$ satisfying

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \tag{1}$$

The conjecture remains open. We use the following classical algebraic reduction; for historical background, see Mordell [1] and Vaughan [2].

Fix $x \in \mathbb{N}_{>0}$ with $4x > n$, and set

$$a = 4x - n, \qquad b = nx.$$

Then

$$\frac{4}{n} - \frac{1}{x} = \frac{a}{b},$$

and multiplication and rearrangement give

$$(ay - b)(az - b) = b^2. \tag{2}$$

Lemma 1.1 (positive divisor split)**.** *Let $n, x, a, b, d_1, d_2, y, z \in \mathbb{N}_{>0}$. Suppose*

$$a = 4x - n, \qquad b = nx, \qquad d_1 d_2 = b^2,$$

and

$$ay = b + d_1, \qquad az = b + d_2.$$

Then (x, y, z) satisfies (1).

Proof. The last three equations are precisely the factorisation (2). Expanding it and using $a = 4x - n$ and $b = nx$ gives

$$4xyz = n(xy + xz + yz).$$

All four variables n, x, y, z are nonzero, so division by $nxyz$ gives (1). $\square$

The Lean development separates these two steps. It first proves the cross-multiplied identity over $\mathbb{Z}$, then proves that a positive instance over $\mathbb{N}$ gives the rational equality in $\mathbb{Q}$.

2 A positive certificate

The offset form avoids subtraction and is convenient both mathematically and formally.

Theorem 2.1 (positive offset certificate)**.** *Let $n, x, p, b, d_1, d_2, y, z \in \mathbb{N}$, with $n, x, p, y, z > 0$. If*

$$4x = n + p, \qquad b = nx, \tag{3}$$

$$d_1 d_2 = b^2, \qquad py = b + d_1, \qquad pz = b + d_2, \tag{4}$$

then (x, y, z) satisfies (1).

Proof. The equations are the positive divisor split with $a = p = 4x - n$. Equivalently, substitution in the cross-multiplied target leaves the multiple

$$p\,(d_1 d_2 - b^2),$$

which vanishes by (4). Positivity then permits passage to the rational identity. $\square$

In Lean, `IsES n x y z` contains both the four positivity assertions and the equality (1) in $\mathbb{Q}$. Theorem 2.1 is the declaration `isES_of_offset_certificate`.

3 The constructive offset theorem

Theorem 3.1 (constructive offset theorem)**.** *Let $n, p, d_1 \in \mathbb{N}$ satisfy*

$$n \geq 2, \qquad p \text{ prime}, \qquad p \equiv 3 \pmod 4, \qquad n \equiv 1 \pmod 4, \qquad p \nmid n.$$

Set

$$x = \frac{n + p}{4}, \qquad b = nx.$$

Suppose $d_1 > 0$, $d_1 \mid b$, and

$$p \mid b + d_1, \tag{5}$$

equivalently $d_1 \equiv -b \pmod p$. Define

$$d_2 = \frac{b^2}{d_1}, \qquad y = \frac{b + d_1}{p}, \qquad z = \frac{b + d_2}{p}. \tag{6}$$

Then $x, y, z \in \mathbb{N}_{>0}$, all divisions in (6) are exact, and (x, y, z) satisfies (1).

Proof. The congruences on n and p show that $4 \mid n + p$, so $x \in \mathbb{N}$ and $4x = n + p$. Moreover $x > 0$. If $p \mid x$, then $p \mid 4x = n + p$, hence $p \mid n$, a contradiction. Thus $p \nmid b = nx$.

Since $d_1 \mid b$, the quotient $d_2 = b^2/d_1$ is exact and positive. It remains to prove $p \mid b + d_2$. Work in the field $\mathbb{Z}/p\mathbb{Z}$. Condition (5) gives

$$d_1 = -b.$$

The element b is nonzero, hence so is d_1. From $d_1 d_2 = b^2$, we may cancel d_1 to obtain

$$d_2 = -b,$$

and therefore $p \mid b + d_2$. Both definitions of y, z in (6) are consequently exact. Their numerators are positive, so $y, z > 0$. Theorem 2.1 now applies. $\square$

Remark. The primality assumption is used only for cancellation modulo p. The Lean proof implements this step in `ZMod p`; the remainder of the construction uses natural numbers, with the final equality stated over the rationals.

4 Fixed-divisor lifting

For fixed p, define

$$X_p(n) = \frac{n+p}{4}, \qquad B_p(n) = nX_p(n),$$

where division is natural-number division. Call a triple (p, d, n) *offset-admissible* if

$$n \geq 2, \quad p \text{ prime}, \quad p \equiv 3 \pmod 4, \quad n \equiv 1 \pmod 4, \quad p \nmid n,$$
$$d > 0, \quad d \mid B_p(n), \quad p \mid B_p(n) + d.$$

These are exactly the hypotheses of Theorem 3.1, packaged with a fixed divisor d.

Theorem 4.1 (fixed-divisor lifting)**.** *If (p, d, n) is offset-admissible, then for every $k \in \mathbb{N}$,*

$$n_k = n + 4pdk$$

is offset-admissible for the same p, d. Consequently, on setting

$$x_k = X_p(n_k), \qquad b_k = B_p(n_k), \qquad d_{2,k} = \frac{b_k^2}{d},$$

the positive natural numbers

$$y_k = \frac{b_k + d}{p}, \qquad z_k = \frac{b_k + d_{2,k}}{p}$$

satisfy

$$\frac{4}{n_k} = \frac{1}{x_k} + \frac{1}{y_k} + \frac{1}{z_k}.$$

Moreover, $k \mapsto n_k$ is strictly increasing, so these parameters form an infinite arithmetic progression.

Proof. The congruences defining admissibility make $n + p$ divisible by 4. For $n_k = n + 4pdk$, exact division gives

$$X_p(n_k) = X_p(n) + pdk. \tag{7}$$

Expanding $B_p(n_k) = n_k X_p(n_k)$ using (7) gives the exact identity

$$B_p(n_k) = B_p(n) + pdk\big(n + 4X_p(n) + 4pdk\big). \tag{8}$$

The added term in (8) is divisible by both d and p. It follows respectively that $d \mid B_p(n_k)$ and $p \mid B_p(n_k) + d$. Also $n_k \equiv n \pmod 4$ and $n_k \equiv n \pmod p$, so the two modular hypotheses are preserved; all other admissibility conditions are unchanged or immediate from $n_k \geq n$. Theorem 3.1 supplies the displayed positive representation, including exactness of all divisions. Finally,

$$n_{k+1} - n_k = 4pd > 0$$

because a prime p and the admissible divisor d are positive. $\square$

5 Explicit infinite families

We record the families formalised in the accompanying Lean file. In every corollary, $k \in \mathbb{N}$.

Corollary 5.1 (Family I)**.** *Let*

$$n = 12k + 9, \qquad x = 3k + 3, \qquad b = n(k + 1).$$

Then

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{b+1} + \frac{1}{b(b+1)}.$$

Proof. Here $4x - n = 3$ and $\gcd(3, nx) = 3$, so the reduced numerator is 1. Taking $d_1 = 1$ and $d_2 = b^2$ in the divisor split gives the displayed denominators. The Lean proof is a direct positive polynomial identity. □

Corollary 5.2 ($p = 3, d_1 = 2$)**.** *Let*

$$n = 24k + 5 \qquad \text{or} \qquad n = 24k + 13.$$

Put $x = (n + 3)/4$ *and* $b = nx$*. Then*

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{(b+2)/3} + \frac{1}{(b+b^2/2)/3}.$$

Proof. The triples $(3, 2, 5)$ and $(3, 2, 13)$ are offset-admissible. Applying Theorem 4.1 gives periods $4 \cdot 3 \cdot 2 = 24$ and hence the two displayed progressions. □

Corollary 5.3 ($p = 3, d_1 = 5$)**.** *Let*

$$n \in \{60k + 5,\ 60k + 17,\ 60k + 25,\ 60k + 37\}.$$

Put $x = (n + 3)/4$ *and* $b = nx$*. Then*

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{(b+5)/3} + \frac{1}{(b+b^2/5)/3}.$$

Proof. The triples with $p = 3, d_1 = 5$ and seeds

$$n \in \{5, 17, 25, 37\}$$

are offset-admissible. Theorem 4.1 gives period $4 \cdot 3 \cdot 5 = 60$, yielding the four displayed progressions. □

The two progressions in Corollary 5.2 are the nonmultiples of 3 in the classical Mordell region $n \equiv 5 \pmod 8$; the remaining progression $n = 24k + 21$ is contained in Family I. The purpose of including them is to exhibit the general construction and its formal specialisation, not to claim a new residue family.

6 Lean formalisation

The formalisation is contained in `lean/ESTheorem.lean` and uses Lean 4 with Mathlib. The complete source is available at https://github.com/carlok/erdos-straus-offset-lean. Its public positive interface consists of:

- `IsES` and `isES_of_polynomial`;

- `isES_of_offset_certificate`;
- `second_divisibility`;
- `constructive_offset`;
- the reducible definitions `offsetX`, `offsetB`, and the packaged interface `OffsetAdmissible` with `OffsetAdmissible.isES`;
- `offsetX_add_period`, `offsetB_add_period`, and `OffsetAdmissible.add_period`;
- `fixed_d_progression` and `fixed_d_progression_strictMono`;
- `family_I_positive`, `family_d2_24k5`, `family_d2_24k13`, and the four `family_d5_60k·` declarations.

For every explicit progression, the file also checks the instances $k = 0$ and $k = 1$. The theorem statement includes the positive denominators and the rational unit-fraction equality; the formalisation is therefore not merely a check of a polynomial identity.

7 Scope

The divisor split is classical, and offset choices of x belong to the standard toolkit for Egyptian-fraction problems. The contribution of this note is a compact theorem interface, an end-to-end Lean proof of the positive construction, and a small collection of formally checked infinite specialisations. The full Erdős–Straus conjecture remains outside its scope.

References

[1] L. J. Mordell, *Diophantine Equations*, Academic Press, 1969.

[2] R. C. Vaughan, *On a problem of Erdős, Straus and Schinzel*, Mathematika 17 (1970), 193–198, doi:10.1112/S0025579300002886.

[3] The Mathlib Community, *Mathlib: the mathematical library of Lean 4*, project repository.